RYAN PACKER

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LinkedIn: linkedin.com/in/ryan-packer-39820 GitHub: github.com/rtpacker/clinical-data-science-portfolio

PROFESSIONAL SUMMARY

Data Scientist & Statistician specializing in healthcare analytics and large-scale EHR data integration, with experience authoring and executing Statistical Analysis Plans (SAPs) for complex observational studies. Expert in architecting reproducible R workflows for high-dimensional clinical data, with professional experience in Python, SQL, and SAS. Proven track record of building automated data pipelines and deploying advanced predictive models, including survival analysis and patient risk stratification. Pursuing AWS Certified Generative AI Developer – Professional to deepen expertise in architecting and deploying scalable, production-grade generative AI and LLM applications on AWS.

EXPERIENCE

UNIVERSITY OF VERMONT LARNER COLLEGE OF MEDICINE — Burlington, VT

Biostatistician, *January 2022 – July 2025*

Multi-Institutional Thrombosis Study (MITH Project) - 3rd co-author, *Journal of Thrombosis & Haemostasias*

- Built a centralized pipeline integrating large scale EHR datasets across six hospital systems including admissions, labs, medications, past medical history and ICD, CPT, HCPCS codes, producing a unified analytic dataset covering 200,000 plus inpatient encounters.
- Standardized complex clinical definitions such as ICU classification and cancer status, resolving cross site data inconsistencies and enabling reliable multi-institution analysis.
- Developed automated and modular data cleaning workflows in R using data.table and dplyr, incorporating robust QC checks to improve reproducibility and reduce manual processing for NIH funded analyses.
- Created time indexed exposure features including pre-event medication windows to support causal modeling of hospital acquired outcomes.

Link: https://doi.org/10.1016/j.jtha.2023.10.015

Heparin-Induced Thrombocytopenia (MITH Sub-study)

- Designed a scalable cohort identification framework combining laboratory results, medication administration data, and diagnostic codes to detect high risk events across more than 100,000 hospitalizations.
- Engineered time-based features such as rolling exposure windows to enable detailed temporal modeling of adverse drug events and treatment patterns.
- Implemented audit ready validation procedures to ensure full reproducibility of cohort logic across institutions and minimize downstream modeling error.

Clonal Hematopoiesis and Stroke Risk in REGARDS - Genomic risk modeling, using national cohort data.

- Integrated high dimensional genomic data with longitudinal clinical outcomes for a national cohort of 30,000 participants to construct features for stroke risk stratification.
- Built predictive survival models using Cox proportional hazards methods and specialized R packages to account for complex sampling designs and reduce selection bias.
- Conducted interaction analyses to quantify joint effects of genomic markers and demographic factors on vascular risk.
- Communicated results through clear visualizations and technical summaries, translating advanced statistical outputs into actionable insights for clinicians and researchers.

CORE TECHNICAL SKILLS

- **Programming:** R (tidyverse, data.table, Shiny); Python (Scikit-learn, pandas, Numpy, RegEx), SQL and SAS.
- **Statistical Modeling and Analytics:** Cox proportional hazards models, logistic regression, mediation analysis, interaction analysis, stratification analysis, model evaluation using AUC and related performance metrics.
- **Visualization** – ggplot2, clinical figures, publication-ready plots
- **Cloud** –AWS (S3, EC2, Bedrock, etc.– training in progress) RAG, LLM.

EDUCATION

UNIVERSITY OF VERMONT, Burlington, VT - M.S. in Statistics

SAINT MICHAEL'S COLLEGE, Colchester, VT- B.S. in Mathematics

CERTIFICATIONS & TRAINING: AWS Certified Solutions Architect Associate - in progress